

MapFormer extensions — consolidated report v2

Supersedes `REPORT.md` (2026-05-15) and `REPORT_ADDENDUM.md` (2026-05-18). Folds in everything since: the full TEM-setting headline (now with 4 generalization axes populated), the cross-scale architecture chain (SingleSize $\rightarrow$ PerScale $\omega \rightarrow$ Hopfield head $\rightarrow$ NoMainAP $\rightarrow$ ExtraHead capacity control), the **corrected capacity verdict** (per-regime + length sweep overturning the earlier Im200-only "CAPACITY" reading), TEM parameter-scaling, NumberLine arithmetic-as-navigation, and the TEMFaithful_FFN direct test.

Every claim cites its source MD; numbers are $n=3$ unless noted.

0. Framing — what changed since REPORT.md

Three things shifted between REPORT.md and now.

- 1. The capacity question got a proper per-regime answer.** The earlier `CAPACITY_CONTROL.md` tested only single-env Im200 at $T=512$ and declared "CAPACITY" because a generic extra attention head matched Level15 on that one point. The corrected control (`CAPACITY_PERREGIME.md`) tests clean / noise / Im200 across T in $\{512, 1024, 2048\}$ plus a NumberLine OOD chain. The verdict is ARCHITECTURE on clean / noise / length / arithmetic / calibration; Im200 alone has a content-channel effect where an extra content head retrieves unique landmark tokens. REPORT.md / earlier writeups oversold "Level15 beats Vanilla on Im200 accuracy" as a generic win — that specific Im200 accuracy number is partly reproducible by a generic extra head, but the architectural claim survives on every other regime that matters.
- 2. The cross-scale story walked through five mechanism candidates and landed on "extra-head capacity, not Hopfield structure."** The chain: TEM dominates small grids $\rightarrow$ coupled- ω hypothesis $\rightarrow$ `SINGLE_SIZE_CONTROL` confirms it $\rightarrow$ per-scale ω architectural fix closes $\sim$ half the gap $\rightarrow$ Hopfield head matches TEM $\rightarrow$ `HOPFIELD_NOMAINAP` ablation shows main-AP is load-bearing $\rightarrow$ `EXTRAHEAD_CONTROL` matches Hopfield with a generic head. The cross-scale cure is more attention capacity / seed-instability rescue, not TEM-shaped memory. TEM remains the most seed-stable.
- 3. The TEM-setting headline (4 axes) is fully populated and defensible.** Held-out-env / cross-topology / cross-scale / cross-class with multi-seed numbers for RoPE, Vanilla, Level15, Level15GSF_NoDrop, Level15GSF_NoDrop_K16 and TEMFaithful. Cognitive- tier architectures all clear the bar; the choice between Level15- family and TEM is regime-dependent. The TEMFaithful_FFN direct test landed — adding a per-position FFN nudges TEM clean from 0.966 $\rightarrow$ 0.969 (not enough to close the lag) but improves NLL meaningfully.

The one-line workshop pitch from REPORT.md still holds:

"MapFormer extends to novel environments along four axes — held-out same-class, cross-topology, cross-scale, cross-class. Across all four, our Level 1.5 correction matches or exceeds TEM- style explicit memory."

with the corrected caveats spelled out in §5 and §11.

1. Models

Baselines and reference architectures

- **MapFormer-WM** — paper-faithful, single-rotation path integrator, position scoring added to attention logits (additive OR-gate).
- **MapFormer-EM** — paper-faithful, separate `q0_pos` / `k0_pos` rotations, `A = softmax(A_X . A_P)` (multiplicative AND-gate).
- **RoPE** — vanilla transformer with rotary position embeddings. Standard-transformer baseline.
- **Vanilla** — Vanilla MapFormer-WM (no Kalman correction). The main non-cognitive baseline. ~256K params.
- **VanillaNoDrop** — Vanilla with the post-attention residual dropout layer removed (see §5). Control for the NoDrop finding.
- **LSTM, CoPE, MambaLike** — non-MapFormer baselines from earlier sessions; numbers in `RESULTS_PAPER.md`.

Correction-augmented MapFormers

- **Level1 / Level1.5 / Level2 InEKF** — Invariant EKF on the path- integrator output `theta_hat`. Level 1 is the parallel-scan steady-state-gain variant; Level 1.5 adds a constant learnable `log_Pi` and a per-token-type `log_R_head` (best speed/accuracy tradeoff); Level 2 adds full per-token heteroscedastic `R_t` (~60× slower training).
- **Level15EM** — Level 1.5 on the EM backbone (`log_R_init_bias=3.0` to prevent the AND-gate init pathology).
- **Level15_PerScaleOmega** — Level 1.5 with one learnable ω per training grid size, selected at forward time by an `env_sizes` kwarg. Targets the coupled- ω cross-scale bottleneck.
- **Level15_Hopfield** — Level 1.5 + a TEM-style position-keyed Hopfield retrieval head (parallel attention head with KV restricted to obs positions, key from position-only rotated `k0_pos`).
- **Level15_Hopfield_NoMainAP** — Level15_Hopfield with the main attention's position rotations replaced by identity. Tests whether position-modulated main attention is load-bearing once the Hopfield head exists.
- **Level15EM_Hopfield** — EM backbone + Hopfield head. (Listed in `EM_HOPFIELD_CROSSSCALE.md` ; Level15EM_PerScaleOmega is the main EM-backbone fix tested.)
- **Vanilla_ExtraHead** — Vanilla + a GENERIC extra attention head (content Q/K, position-modulated, all-positions causal KV). The capacity control. 322K params (more than Level15).
- **Level15_ExtraHead** — Level 1.5 + same generic extra head. The cross-scale capacity control.
- **L15_NoCorr** — Level 1.5 architecture (305K params) with the InEKF correction zeroed at runtime. Same params as Level15, no correction. Param-matched control.
- **Level15NoDrop** — Level 1.5 with the inherited post-attention residual dropout removed.
- **Level15GSF** — Gaussian Sum Filter with K=8 parallel Level 1.5 chains; learnable per-chain `theta_init_k` offsets; mixture weights from cumulative log-likelihood.
- **Level15GSF_NoDrop / _NoDrop_K16** — GSF stacked with NoDrop; K16 variant uses 16 modes.

TEM family

- **TEMFaithful** — Whittington-style TEM with separate g (structural code) and Hopfield memory bank keyed by g , $W_a = \exp(\text{skew}(A_a))$ orthogonal per-action rotation in d_g space, predict-then-update. The bug-fixed version (queries memory with POST-action g).
- **TEMFaithful_FFN** — TEMFaithful + a per-position FFN on the retrieved content (fixed Hopfield bank unchanged).
- **TEMFaithful_dg{32, 64, 128, 256}** — parameter sweep on d_g . The $d_g=64$ case is the default; $d_g=256$ is parameter-matched to Level15.
- **TEM-T** — transformer-formulation of TEM; reference comparison only.

NumberLine variants

- **NumberLine{Vanilla, Level15, Vanilla_ExtraHead}** — MapFormer variants on a 1D additive torus ($N=64$, 6 ops). Tests whether the Level15 self-correcting accumulator extrapolates to longer arithmetic chains.

2. Headline — TEM-setting novel-environment generalization

Source: TEM_NOVEL_ENV_RESULTS.md , MULTIENV_CLEAN_2x2.md , MULTICLASS_MULTISEED_RESULTS.md , MULTISEED_FOLLOWUP_RESULTS.md , TEM_BACKGROUND_BASELINES.md . All $n=3$.

2.1 Multi-env held-out (TEM's classic novel-env test)

50 train envs, 50 held-out test envs, size 64, $T=128$ train.

LM200 (200 landmarks):

Variant	Train	Held T=128	Held T=512 OOD
RoPE	0.587 ± 0.024	0.595 ± 0.027	0.503 ± 0.003
Vanilla	0.823 ± 0.039	0.820 ± 0.044	0.728 ± 0.047
Level15	0.997 ± 0.001	0.997 ± 0.002	0.988 ± 0.003
Level15GSF_NoDrop	0.997 ± 0.002	0.996 ± 0.002	0.976 ± 0.005
TEMFaithful	1.000 ± 0.000	1.000 ± 0.000	0.967 ± 0.004

CLEAN (no landmarks):

Variant	Train	Held T=128	Held T=512 OOD
RoPE	0.589 ± 0.080	0.592 ± 0.066	0.503 ± 0.018
Vanilla	0.994 ± 0.005	0.992 ± 0.007	0.920 ± 0.024
Level15	0.998 ± 0.003	0.998 ± 0.003	0.975 ± 0.010
Level15GSF_NoDrop	1.000 ± 0.000	1.000 ± 0.000	0.989 ± 0.006
TEMFaithful	1.000 ± 0.000	1.000 ± 0.000	0.976 ± 0.002 ($n=2$)

Cognitive-tier all in 0.97-0.99 OOD. Standard transformers are not. Choice between Level15-family and TEM is regime-dependent (TEM slightly better with landmarks, Level15GSF_NoDrop slightly better on clean).

2.2 Cross-topology (TEM OOD-d analog)

Train on torus + open + walls mix; eval per-topology on held-out envs at T=512 OOD.

Variant	torus	open	walls
RoPE	0.468 $\pm$ 0.027	0.510 $\pm$ 0.027	0.536 $\pm$ 0.006
Vanilla	0.787 $\pm$ 0.039	0.717 $\pm$ 0.032	0.658 $\pm$ 0.038
Level15	0.907 $\pm$ 0.065	0.818 $\pm$ 0.050	0.748 $\pm$ 0.029
Level15GSF_NoDrop	0.926 $\pm$ 0.043	0.835 $\pm$ 0.026	0.788 $\pm$ 0.009
Level15GSF_NoDrop_K16	0.955 $\pm$ 0.002	0.855 $\pm$ 0.002	0.778 $\pm$ 0.002
TEMFaithful	0.907 $\pm$ 0.002	0.823 $\pm$ 0.003	0.796 $\pm$ 0.002

2.3 Cross-scale (TEM OOD-s analog)

Train on sizes 32 / 64 / 128; eval per-size at T=512 OOD.

Variant	size 32	size 64	size 128
RoPE	0.396 $\pm$ 0.040	0.492 $\pm$ 0.056	0.506 $\pm$ 0.051
Vanilla	0.535 $\pm$ 0.043	0.719 $\pm$ 0.066	0.747 $\pm$ 0.075
Level15	0.782 $\pm$ 0.138	0.921 $\pm$ 0.050	0.953 $\pm$ 0.032
Level15GSF_NoDrop	0.838 $\pm$ 0.111	0.947 $\pm$ 0.034	0.967 $\pm$ 0.022
Level15GSF_NoDrop_K16	0.826 $\pm$ 0.109	0.951 $\pm$ 0.031	0.973 $\pm$ 0.019
TEMFaithful	0.936 $\pm$ 0.021	0.973 $\pm$ 0.006	0.981 $\pm$ 0.005

TEM dominates small grids by +11pp with much tighter seed variance (0.021 vs 0.109). §3 walks through the mechanism investigation.

2.4 Cross-class (beyond TEM): torus + MiniGrid-DoorKey

Different action vocab (4 vs 7), different obs spaces, env-prefix tokens.

Variant	Torus T=128	Torus T=512	DoorKey T=128	DoorKey T=512
RoPE	0.548 $\pm$ 0.030	0.497 $\pm$ 0.008	0.890 $\pm$ 0.011	0.788 $\pm$ 0.012
Vanilla	0.813 $\pm$ 0.039	0.681 $\pm$ 0.091	0.957 $\pm$ 0.008	0.841 $\pm$ 0.022
Level15	0.925 $\pm$ 0.024	0.879 $\pm$ 0.039	0.946 $\pm$ 0.003	0.888 $\pm$ 0.010
Level15GSF_NoDrop_K16	0.925 $\pm$ 0.020	0.865 $\pm$ 0.032	0.945 $\pm$ 0.002	0.891 $\pm$ 0.000
TEMFaithful	n/a	n/a	n/a	n/a

TEMFaithful's `tokens<n_actions` action/obs split doesn't accommodate the unified vocab (env-prefix tokens at IDs 0/1 + actions spread over 2-12). Would need a custom is-action mask to compare.

3. Cross-scale investigation — five candidates, mechanism walked

The +11pp small-grid TEM gap (§2.3) prompted a sequence of falsifiable tests. Each row is the "winner-so-far" hypothesis; subsequent rows either confirm or falsify it.

3.1 TEM cross-scale diagnostic (analytic)

Source: `TEM_CROSSSCALE_DIAGNOSTIC.md`. Four candidate mechanisms: TEM's `w_a` is scale-invariant by construction; Level15's path integrator has scale-coupled ω ; TEM's Hopfield retrieval is scale-agnostic; GSF mode init is scale-coarse. The cheapest test: per-scale ω . The most informative: single-size Level15.

3.2 SingleSize control — coupled- ω confirmed

Source: `SINGLE_SIZE_CONTROL.md`. Train Level15 lm200 on a SINGLE grid size; compare to multi-size mix.

Variant	size 16 T=512	size 32 T=512
Level15 single-size	0.159 ± 0.019	0.908 ± 0.085
Level15 multi-size (32+64+128)	—	0.782 ± 0.138
TEMFaithful multi-size (ref)	—	0.936 ± 0.021

Single-size 32 (0.908) >> multi-size 32 (0.782) → **+13pp**. Closes most of the gap to TEM. Coupled- ω confirmed as a real bottleneck. Size 16 collapses for all ($n_{\text{blocks}}=32 > 256$ positions; different failure mode).

3.3 Per-scale ω — partial fix

Source: `PERSCALE_OMEGA_RESULTS.md`. One learnable ω per training scale, selected by `env_sizes` kwarg.

Variant	size 32	size 64	size 128
Level15 (coupled ω)	0.782 ± 0.138	0.921 ± 0.050	0.953 ± 0.032
Level15_PerScaleOmega	0.877 ± 0.163	0.937 ± 0.075	0.946 ± 0.074
TEMFaithful (ref)	0.936 ± 0.021	0.973 ± 0.006	0.981 ± 0.005

Size 32: +10pp over coupled- ω Level15. Closes about half the gap to TEM. Sizes 64/128 essentially tied. High variance (± 0.163) — at least one seed under-converged. Real architectural improvement at small grids, but not the full story.

3.4 Hopfield head — closes the gap

Source: `EM_HOPFIELD_CROSSSCALE.md`. Add a TEM-style position-keyed Hopfield retrieval head to Level15 / Level15EM.

Variant	size 32	size 64	size 128
Vanilla	0.535 ± 0.043	0.719 ± 0.066	0.747 ± 0.075
Level15	0.782 ± 0.138	0.921 ± 0.050	0.953 ± 0.032
Level15EM	0.740 ± 0.042	0.887 ± 0.051	0.913 ± 0.060
Level15_PerScaleOmega	0.877 ± 0.163	0.937 ± 0.075	0.946 ± 0.074

Variant	size 32	size 64	size 128
Level15EM_PerScaleOmega	0.766 ± 0.063	0.930 ± 0.032	0.956 ± 0.028
Level15_Hopfield	0.919 ± 0.089	0.972 ± 0.030	0.985 ± 0.018
TEMFaithful (ref)	0.936 ± 0.021	0.973 ± 0.006	0.981 ± 0.005

Level15_Hopfield matches TEMFaithful within seed noise at every scale. First read: "we ported TEM's memory mechanism into MapFormer at parallel cost." Spoiler: §3.6 falsifies that framing.

Level15EM consistently underperforms Level15 at every scale — backbone choice not orthogonal to small-grid issue.

3.5 NoMainAP — main attention's position channel IS load-bearing

Source: `HOPFIELD_NOMAINAP_RESULTS.md`. Identity-rotate the main attention's position channel; position enters the model ONLY through the Hopfield head.

Variant	size 32	size 64	size 128
Level15	0.782 ± 0.138	0.921 ± 0.050	0.953 ± 0.032
Level15_Hopfield	0.919 ± 0.089	0.972 ± 0.030	0.985 ± 0.018
Level15_Hopfield_NoMainAP	0.627 ± 0.189	0.741 ± 0.188	0.750 ± 0.196
TEMFaithful (ref)	0.936 ± 0.021	0.973 ± 0.006	0.981 ± 0.005

Removing main-AP **collapses** Level15_Hopfield. The factored (TEM- style) design is NOT sufficient; MapFormer's position-modulated main attention is load-bearing. The Hopfield head is a supplement, not a replacement. So whatever the Hopfield head is doing, it's not substituting for main-AP — it's adding something on top.

3.6 ExtraHead capacity control — the Hopfield win is capacity

Source: `EXTRAHEAD_CONTROL.md`. Replace the Hopfield head with a GENERIC extra attention head (content Q/K, position-modulated, all- positions causal KV) — same residual wiring, **more parameters**.

Variant	size 32	size 64	size 128
Level15	0.782 ± 0.138	0.921 ± 0.050	0.953 ± 0.032
Level15_Hopfield	0.919 ± 0.089	0.972 ± 0.030	0.985 ± 0.018
Level15_ExtraHead	0.934 ± 0.081	0.979 ± 0.028	0.984 ± 0.022
TEMFaithful (ref)	0.936 ± 0.021	0.973 ± 0.006	0.981 ± 0.005

Per-seed at size 32:

Variant	seed 0	seed 1	seed 2
Level15	0.977	0.701	0.668
Level15_Hopfield	0.986	0.793	0.977
Level15_ExtraHead	0.987	0.819	0.996

A generic extra head matches the Hopfield head (0.934 vs 0.919, indistinguishable at this variance). Base Level15 collapses on 2 of 3 seeds at size 32 — the coupled- ω bad basin.

Adding ANY extra attention head rescues most seeds; the position-only-key / obs-restricted-KV structure is not necessary.

The earlier "we ported TEM's memory mechanism" framing is NOT supported. The cross-scale fix is capacity / training stabilisation, not Hopfield structure.

TEMFaithful remains the most seed-stable (± 0.021 vs ± 0.081). The extra-head fix reduces but does not eliminate Level15's seed instability.

4. Controls — what does and does not generalise

4.1 VanillaNoDrop — InEKF correction is doing real work

Source: `VANILLANODROP_CONTROL.md` . Multi-env held-out, $n=3$.

Setting	Vanilla	VanillaNoDrop	Level15
LM200 T=512 OOD	0.728 ± 0.047	0.737 ± 0.028	0.988 ± 0.003
CLEAN T=512 OOD	0.920 ± 0.024	0.962 ± 0.020	0.975 ± 0.010

VanillaNoDrop $\approx$ Vanilla on lm200, $\sim +4$ pp on clean. **The Level15NoDrop +13pp single-env lm200 win is NOT reducible to a dropout fix — InEKF correction is doing real work.** Workshop-critical control.

4.2 Capacity question — corrected verdict (CRITICAL)

Source: `CAPACITY_PERREGIME.md` . Supersedes `CAPACITY_CONTROL.md` .

The earlier control tested only single-env lm200 at T=512 and saw `Vanilla_ExtraHead` (0.875) $\geq$ `Level15` (0.828). Verdict was "CAPACITY" — Level15's win came from its extra params, not its architecture.

The per-regime control overturns this. Per-regime (T=512 OOD, $n=3$):

Regime	Vanilla	Vanilla_ExtraHead	Level15	verdict
clean	0.918 ± 0.042	0.742 ± 0.049	0.995 ± 0.004	ARCHITECTURE
noise	0.634 ± 0.031	0.648 ± 0.034	0.707 ± 0.012	ARCHITECTURE
lm200	0.721 ± 0.054	0.875 ± 0.139	0.828 ± 0.023	CAPACITY-looking

On clean, the extra head is *worse* than Vanilla (the extra content attention destabilises the path-integration regime). On noise, the extra head matches Vanilla — doesn't reach Level15. Only lm200 reaches Level15, with high variance (± 0.139).

Length sweep — clean accuracy. The bounded-error signature is visible only across lengths:

Variant	T=512	T=1024	T=2048
Vanilla	0.918 ± 0.042	0.802 ± 0.062	0.627 ± 0.035
Vanilla_ExtraHead	0.742 ± 0.049	0.601 ± 0.026	0.526 ± 0.018
Level15	0.995 ± 0.004	0.968 ± 0.006	0.886 ± 0.013

Gap at T=512: +8pp. Gap at T=2048: **+26pp**. Capacity is the worst at every length; only the wrap-stabilised filter holds up.

Length sweep — clean NLL (calibration):

Variant	T=512	T=1024	T=2048
Vanilla	0.410 ± 0.162	1.352 ± 0.424	3.084 ± 0.625

Variant	T=512	T=1024	T=2048
Vanilla_ExtraHead	1.620 $\pm$ 0.209	3.023 $\pm$ 0.191	3.978 $\pm$ 0.363
Level15	0.031 $\pm$ 0.023	0.178 $\pm$ 0.050	0.648 $\pm$ 0.084

Level15 is 5-50 $\times$ better than both. The extra head makes NLL *worse* across the board.

NumberLine — arithmetic OOD chain (T=512, 4 $\times$ trained chain length):

Variant	in-dist T=128	OOD chain T=512	T=512 NLL
Vanilla	0.925 $\pm$ 0.009	0.633 $\pm$ 0.071	2.024 $\pm$ 0.666
Vanilla_ExtraHead	0.986 $\pm$ 0.000	0.662 $\pm$ 0.130	2.542 $\pm$ 1.324
Level15	0.902 $\pm$ 0.023	0.841 $\pm$ 0.056	0.521 $\pm$ 0.292

Capacity (Vanilla_ExtraHead) does NOT close Level15's +21pp arithmetic- extrapolation gap.

Honest verdict:

- **ARCHITECTURE** on clean accuracy, noise accuracy, length generalization (clean / noise / lm200), arithmetic extrapolation, and calibration (NLL) at every regime / length.
- **CAPACITY-looking** on single-env lm200 accuracy: a generic extra *content* head can find unique landmark tokens with content-only attention. The mechanism is content retrieval of one-shot tokens, not generic param-count.

The corrected claim: Level15's win on the regimes that matter (clean OOD, length extrapolation, calibration, arithmetic) is architectural — it's the wrap + per-token-type gating doing the work, not 50K extra parameters. The single-env lm200 accuracy headline specifically is partly reproducible by an extra content head, which should be flagged honestly.

4.3 L15_NoCorr — param-matched no-correction control

Same table (CAPACITY_PERREGIME.md / CAPACITY_CONTROL.md). L15_NoCorr = Level15 architecture (305K params) with the lnEKF correction zeroed at runtime. lm200 T=512 OOD: 0.677 $\pm$ 0.001 — close to Vanilla (0.721), well below Level15 (0.828). Confirms the correction itself (not the extra params) is what's lifting Level15 above Vanilla on lm200 once you control for capacity.

4.4 SingleSize — coupled- ω confirmed (replayed from §3.2)

Already covered above. Listed here as the cross-scale control. The multi-size training regime hurts small grids by ~13pp via shared ω .

5. NoDrop and GSF — the dropout finding and multi-modal Bayes

5.1 NoDrop Pareto-shift (single-env, n=3)

Source: NODROP_PARETO_RESULTS.md .

Config	Vanilla	Level15	Level15NoDrop
Clean T=512	0.911	0.993	0.985
Clean T=512 NLL	0.458	0.039	0.070
Noise T=512	0.638	0.702	0.699

Config	Vanilla	Level15	Level15NoDrop
LM200 T=512	0.716	0.819	0.948
LM200 T=512 NLL	1.391	0.897	0.317

NoDrop is essentially Pareto-equivalent on clean/noise (differences within seed std) and a +13pp win on single-env lm200. Near-free engineering Pareto-shift for landmark regimes.

5.2 NoDrop vs GSF — substitutes on accuracy, complements on NLL

Source: `GSF_NODROP_RESULTS.md`. Single-env lm200, T=512 OOD, n=3.

Variant	Accuracy	NLL
Level15	0.819 ± 0.025	0.897
Level15NoDrop	0.948 ± 0.025	0.317
Level15GSF	0.956 ± 0.042	0.227
Level15GSF_NoDrop	0.961 ± 0.038	0.177
TEMFaithful (ref)	0.969 ± 0.010	0.171

NoDrop and GSF are accuracy-substitutes (+13pp vs +14pp; stacked still +14pp) but **NLL-complements** (stacked 5× better NLL). GSF earns its keep only on calibration / posterior shape, not on raw accuracy. ~4× compute cost. For minimal sweeps, use NoDrop.

5.3 Mechanism — Vaswani's post-attn residual dropout is the culprit

`WMTransformerLayer` (the paper-faithful layer) wraps `o_proj(out)` in `self.dropout` on the residual path. With lots of redundant retrievals (aliased obs: ~128 copies per token type) this regularises; with rare retrievals (a landmark token appearing once) it destroys the signal. Removing the dropout: +13pp on lm200, ~free elsewhere. The "learnable beta / sharper softmax" framing tested in `LEVEL15BETA_RESULTS.md` was a red herring — beta barely moves from init, but `Level15NoDrop` (beta fixed, only dropout removed) matches `Level15Beta` exactly.

6. NumberLine — arithmetic as navigation

Source: `NUMBERLINE_RESULTS.md` (plus the capacity row in §4.2).

MapFormer on a 1D additive torus (N=64, 6 ops: $\pm 1/\pm 2/\pm 3$). The path integrator $\theta = \omega * \text{cumsum}(f_Delta(action))$ literally computes $(a+b+c+...) \bmod N$. Task: predict the obs token at a REVISITED value.

Train chain = 128 ops; OOD chain = 512 ops (4× longer).

Variant	in-dist T=128	OOD chain T=512	T=512 NLL
Vanilla	0.925 ± 0.009	0.633 ± 0.071	2.024 ± 0.666
Vanilla_ExtraHead	0.986 ± 0.000	0.662 ± 0.130	2.542 ± 1.324
Level15	0.902 ± 0.023	0.841 ± 0.056	0.521 ± 0.292

Reading: Level15 +21pp on OOD arithmetic chain at 4× trained length. The self-correcting accumulator generalises to longer chains where uncorrected cumsum drifts. Capacity (Vanilla_ExtraHead, 0.662) does NOT close the gap — gives at most +3pp over Vanilla. NLL gap even larger (Level15 4× lower).

Both in-distribution accuracies are low (~ 0.92) — the aliased-obs recall task on a number line is hard; revisit-prediction needs work. But the arithmetic-extrapolation signal is clean.

7. TEM analysis

7.1 TEM parameter-scaling — saturates, doesn't scale

Source: `TEM_SCALING_RESULTS.md`. TEMFaithful at d_g in {32, 64, 128, 256}. A_a (per-action transition matrices, the bulk of TEM's learnable "model") scales as $4 * d_g^2$. $d_g=256$ ($\sim 291K$ params) is parameter-matched to Level15 ($\sim 305K$).

Variant	d_g	params	T=128	T=512 OOD	T=512 NLL
TEMFaithful_dg32	32	32,766	1.000	0.979 $\pm$ 0.002	0.141 $\pm$ 0.013
TEMFaithful	64	45,086	1.000	0.973 $\pm$ 0.010	0.145 $\pm$ 0.067
TEMFaithful_dg128	128	94,302	1.000	0.966 $\pm$ 0.006	0.192 $\pm$ 0.042
TEMFaithful_dg256	256	291,038	1.000	0.953 $\pm$ 0.008	0.259 $\pm$ 0.043

TEM anti-scales: dg32 (0.979) > dg256 (0.953) on accuracy, and the NLL pattern is monotone in the wrong direction. Strong evidence for the inductive-bias account — TEM is parameter-efficient because the hard part (associative memory via fixed Hopfield retrieval) is algorithmic, not learned. Adding capacity to A_a doesn't help; it hurts.

7.2 TEMFaithful_FFN — does the missing FFN close TEM's clean lag?

Source: `TEM_NOISE_FFN_RESULTS.md`. TEMFaithful + per-position FFN on the retrieved content. Targets the 3pp clean-regime gap vs Level15.

Single-env clean:

Variant	T=128	T=512 OOD	T=512 NLL
TEMFaithful (no FFN)	1.000	0.966 $\pm$ 0.008	0.182
TEMFaithful_FFN	1.000	0.969 $\pm$ 0.002	0.145 $\pm$ 0.003
Level15 (ref)	1.000	0.993	0.039

Adding the FFN nudges TEM clean accuracy from 0.966 $\rightarrow$ 0.969 (within noise — not enough to close the 2.4pp lag to Level15) and improves NLL 0.182 $\rightarrow$ 0.145. The missing-FFN hypothesis is largely falsified: the clean lag is from something else (learned content attention, not per-position processing).

Single-env lm200:

Variant	T=128	T=512 OOD	T=512 NLL
TEMFaithful (no FFN)	1.000	0.969 $\pm$ 0.010	0.171
TEMFaithful_FFN	1.000	0.979 $\pm$ 0.005	0.093 $\pm$ 0.027

Modest +1pp accuracy, NLL nearly halved. Not free machinery but doesn't hurt.

7.3 TEMFaithful in the noise regime

Source: `TEM_NOISE_FFN_RESULTS.md`.

Variant	T=128 noise	T=512 noise OOD	T=512 NLL
TEMFaithful	0.762 ± 0.003	0.709 ± 0.002	1.216 ± 0.010
Vanilla (ref)	—	0.638	—
Level15 (ref)	—	0.702	—
Level15NoDrop (ref)	—	0.699	—

TEMFaithful in the noise regime narrowly beats Level15 / Level15NoDrop on accuracy at T=512 (+0.7pp / +1.0pp) but loses on NLL by a lot (1.216 vs 0.952 for Level15). Tied within practical significance.

8. Other regimes (largely unchanged from REPORT.md)

8.1 Long-T evaluation

Source: `LONGT_EVAL_RESULTS.md`, `CAPACITY_PERREGIME.md` length-sweep tables (§4.2 above). T up to 2048, no retraining. RoPE → MapFormer gap grows with T. Level15's NLL stays low at T=2048; Vanilla's NLL doubles. Level15 clean: 0.886 at T=2048 vs Vanilla 0.627 (+26pp).

8.2 MiniGrid

- DoorKey-8x8 (`MINIGRID_DOORKEY_*.md`, `MINIGRID_DK16_RESULTS.md`): Level15 +10pp noise OOD at T=512, ties clean. Long-T T=2048: noise gap +16pp.
- MemoryS13 (`MINIGRID_MEMORY_RESULTS.md`): cleanest "Level15 wins clean OOD on a real env": +13pp at T=512, +13pp at T=1024, NLL 5× better.

8.3 Vocab sweep

Source: `VOCAB_SWEEP_RESULTS.md`. Single seed, T=512 OOD on fresh obs_map. At n_obs=256, VanillaEM collapses (0.562, worse than Vanilla 0.665). Correction rescues both backbones to ~0.97. All variants collapse at n_obs=4096 (degenerate regime). Paper's "EM wins at large vocab" claim is l=16-specific.

8.4 Sparse landmarks

Source: `SPARSE_LANDMARKS_RESULTS.md`. TEM dominates sparse-landmark density. Frame as a TEM strength; Level15 stays within a few pp.

8.5 Stochastic-transition vs action-record corruption

Source: `STOCHASTIC_TRANSITION_RESULTS.md`. Empirically confirms the equivalence for uniform policies. Small (~5pp) asymmetry; use stochastic-transition vocabulary for the writeup.

9. Mechanism findings (unchanged from REPORT.md)

9.1 Kalman's win is stabilisation + token-type gating

R_t learns to be HIGH on aliased obs (measurement contribution tiny) yet Level 1.5 still beats Vanilla. The structural win comes from: (1) the wrap (atan2 of innovation) — keeps theta_hat

bounded at OOD length; (2) per-token R_t — gates by token TYPE (action vs obs). The wrap is load-bearing for length generalization; gating for clean-task quality.

9.2 EM vs WM is regime-dependent

Regime	A_X	A_P	Winner	Confirmed
Paper aliased + short	noisy	sharp	EM	paper
Aliased + long OOD	noisy	drift-degraded	WM	ours
Landmarks	sharp	drift-degraded	WM	ours
With correction	—	repaired	tied	ours
DoorKey egocentric	very noisy	sharp	EM	DOORKEY_BC_RESULTS.md

9.3 PC and InEKF are mathematical duals

Coupling them creates a degenerate optimum (R-saturation autoencoder bypass). Only full gradient isolation (Level15PC_v4) avoids it, at which point PC adds essentially nothing. See [feedback_pc_kalman_duality.md](#).

10. Goal-directed / behavioural (unchanged from REPORT.md)

Summary numbers below; full detail in REPORT.md §4.

- **Open-loop match-acc** ([GOAL_DIRECTED_RESULTS.md](#)): Level15 +18pp over Vanilla at OOD explore length. Correction-stabilised maps stay navigable.
- **Frozen-probe** ([PROBE_GOAL_RESULTS.md](#)): Level15 +7.5pp over Vanilla on held-out probe (linear readout → action). Cognitive maps differ in CONTENT, not just trainability.
- **Closed-loop goal navigation** ([GOAL_CLOSEDLOOP_RESULTS.md](#)): 1-2% success for everyone. BC distribution shift dominates. Cannot lead the workshop pitch with this.
- **DoorKey BC + DAgger** ([DOORKEY_BC_RESULTS.md](#) , [DAGGER_RESULTS.md](#)): EM wins match-acc on DoorKey (egocentric → A_X noisy → multiplicative gate wins, opposite of torus). DAgger: only Level15NoDrop shows a clear gain (0.24 → 0.42).
- **Active-inference one-step / multi-step** ([ACTIVE_INFERENCE_RESULTS.md](#)): 0-2% across all variants and horizons. The frozen forward model is structurally too myopic to drive multi-step nav.
- **Successor-rep aux pretraining** ([SR_PRETRAIN_RESULTS.md](#)): chance on both goal-distance probe and active-inference closed-loop. Aux loss with a separate head doesn't force the backbone to change.

11. Probes and honest negatives

Refer to REPORT.md §5-§7 plus RESULTS_INDEX.md for the full list. Briefly:

- **Goal-distance probe (head state)** at chance for all variants; TEMFaithful is the only one showing rank-order signal (Spearman 0.27).
- **Position-decode probe** half-chance on displacement-from-start; cognitive map IS there in $\theta_{\hat{}}$, just not at single-cell precision via a 2-layer MLP head.

- **Place cells emerge everywhere; hex doesn't emerge anywhere** (Grid, Grid_Free, Level15_DoG, GridL15PC_Free).
- **Sorscher's three conditions** (path integration + non-negativity
- DoG-similarity targets) not sufficient on the discrete-cell torus.
- **PC + Kalman** architecturally guarantees a degenerate optimum (R-saturation autoencoder bypass).
- **TEM cross-scale dominates at small grids** — addressable by extra attention capacity, not by Hopfield-style structure (§3).
- **TEM parameter-scaling anti-scales** (§7.1).

12. Bottom line / workshop framing

What's defensible (multi-seed, on the regimes that matter):

1. **Cognitive-map architectures (Level15-family / TEM) generalise to novel environments along four axes; standard transformers don't.** Multi-env, cross-topology, cross-scale, cross-class (§2).
2. **Explicit state correction (Level 1.5) is architectural, not capacity.** Per-regime + length sweep + arithmetic OOD all show ARCHITECTURE; only single-env lm200 accuracy is reproducible by adding a generic extra content head (§4.2). The bounded-error signature shows up only across lengths (clean +8pp at T=512 → +26pp at T=2048) and only Level15 has it.
3. **Calibration (NLL) is where Level1.5 dominates most clearly.** 5–50× better than Vanilla / Vanilla_ExtraHead across every regime / length. Capacity *worsens* calibration.
4. **NoDrop is a near-free engineering Pareto-shift; GSF earns its keep on calibration, not accuracy.** §5.
5. **Cognitive maps differ in representation content, not just trainability.** Frozen linear probe +7.5pp Level15 → Vanilla (§10).
6. **Level1.5 extrapolates arithmetic chains.** +21pp on NumberLine OOD chain at 4× trained length; capacity does not close this (§6).
7. **TEM is parameter-efficient by inductive bias, not by under-parameterisation.** TEM anti-scales with d_g (§7.1).
8. **EM vs WM is regime-dependent; mechanism predicts both signs.** Same architecture wins different regimes (§9.2).

What's NOT defensible:

- "Level1.5 is universally best on small grids." TEM wins by +11pp; the cure is extra attention capacity, not Hopfield structure (§3).
- "Closed-loop goal navigation works." False for all variants (§10).
- "Hex grid cells emerge." False everywhere (§11).
- "Single-env lm200 accuracy is purely architectural." A generic extra content head matches Level15 here. Specific to this regime; doesn't generalise (§4.2).

Corrected one-line workshop pitch:

"Explicit state correction (Level 1.5) in MapFormer buys bounded- error length-extrapolation, calibration, and training stability across novel-environment regimes (held-out maps, topologies, scales, task classes) — and matches TEM-style explicit memory on the regimes that matter. On aliased-observation data, generic extra capacity actively *worsens* clean accuracy and calibration. The architectural win survives every per-regime control

except single- env lm200 accuracy, where a generic content head can also retrieve unique landmark tokens."

13. What's in flight

- `run_tem_noise_and_ffn.sh` → `TEM_NOISE_FFN_RESULTS.md` — already landed (§7.2-7.3).
Net: FFN doesn't close clean lag (~2.4pp); lm200 improves; noise regime TEM matches Level15 on accuracy, loses on NLL.

Nothing else queued at the time of writing. All multi-seed cells in §2 and §3 are populated; the per-regime capacity sweep is complete; the TEM scaling sweep is complete.

Appendix — source file index

Headline tables — `TEM_NOVEL_ENV_RESULTS.md`, `MULTIENV_CLEAN_2x2.md`, `MULTICLASS_MULTISEED_RESULTS.md`, `MULTISEED_FOLLOWUP_RESULTS.md`, `TEM_BACKGROUND_BASELINES.md`.

Cross-scale chain — `TEM_CROSSSCALE_DIAGNOSTIC.md`, `SINGLE_SIZE_CONTROL.md`, `PERSCALE_OMEGA_RESULTS.md`, `EM_HOPFIELD_CROSSSCALE.md`, `HOPFIELD_NOMAINAP_RESULTS.md`, `LEVEL15EM_CROSSSCALE.md`, `EXTRAHEAD_CONTROL.md`.

Capacity controls (read **CAPACITY_PERREGIME, not **CAPACITY_CONTROL**)** — `CAPACITY_PERREGIME.md` (controlling), `CAPACITY_CONTROL.md` (superseded — lm200-only artifact), `VANILLANODROP_CONTROL.md`.

NoDrop / GSF — `NODROP_PARETO_RESULTS.md`, `GSF_NODROP_RESULTS.md`, `DROPOUT_ABLATION_RESULTS.md`, `LEVEL15BETA_RESULTS.md`.

TEM analysis — `TEM_SCALING_RESULTS.md`, `TEM_NOISE_FFN_RESULTS.md`, `TEM_T_MULTISEED.md`.

NumberLine — `NUMBERLINE_RESULTS.md` (+ capacity row in `CAPACITY_PERREGIME.md`).

Other regimes — `LONGT_EVAL_RESULTS.md`, `VOCAB_SWEEP_RESULTS.md`, `SPARSE_LANDMARKS_RESULTS.md`, `STOCHASTIC_TRANSITION_RESULTS.md`, `MINIGRID_DOORKEY_*.md`, `MINIGRID_DK16_RESULTS.md`, `MINIGRID_MEMORY_RESULTS.md`.

Behaviour and probes — `GOAL_DIRECTED_RESULTS.md`, `GOAL_CLOSEDLOOP_RESULTS.md`, `PROBE_GOAL_RESULTS.md`, `PROBE_GOAL_DISTANCE.md`, `STATE_PROBES.md`, `DOORKEY_BC_RESULTS.md`, `DAGGER_RESULTS.md`, `ACTIVE_INFERENCE_RESULTS.md`, `SR_PRETRAIN_RESULTS.md`.

Mechanism / feedback notes — `.claude-memory/feedback_post_attn_dropout.md`, `.claude-memory/feedback_em_vs_wm_mechanism.md`, `.claude-memory/feedback_pc_kalman_duality.md`, `.claude-memory/feedback_action_noise_framing.md`.

Older reports superseded by this one — `REPORT.md` (2026-05-15), `REPORT_ADDENDUM.md` (2026-05-18).

Generated 2026-06-02. Verify against `git log` for newer results.